Adversarial Robustness without Training Data
Robustness without data via denoised smoothing and test-time self-adaptation

Daewon Choi, Jongheon Jeong, Huiwon Jang, Jinwoo Shin
KAIST, Korea University

02

Why data-free robustness?

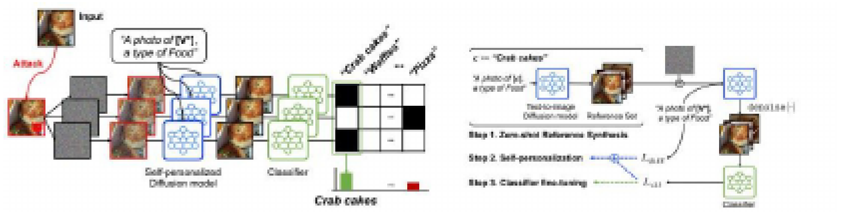
Overview and Motivations

- Goal: boost adversarial robustness without training data
- Threat: adversarial perturbations that shift predictions
- Reality: off-the-shelf models are brittle; data inaccessible
  - Limitation: existing defenses need data and special training
  - Aim: scalable to diverse off-the-shelf vision classifiers
- Framework: use text-to-image diffusion models as denoisers

Sec II

Proposed Framework

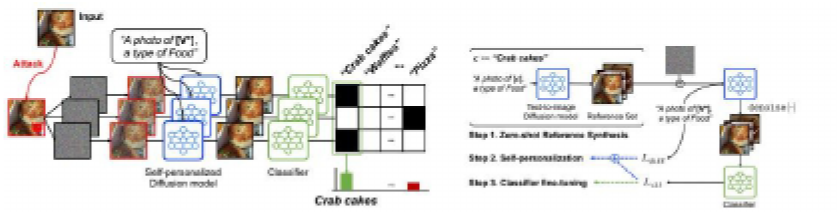
5 Denoised Smoothing Pipeline



(a) Denoised smoothing from text-to-image models (b) Self-adaptation schemes

- Apply input transforms, then add Gaussian noise.
  - Denoise via text-to-image diffusion models.
  - Classify multiple stochastic reconstructions.
- Aggregate votes to yield a smoothed decision.
  - Pixel-space super-resolution denoisers preferred.
  - No training data required for robust gains.

6 Self-adaptation at Test Time



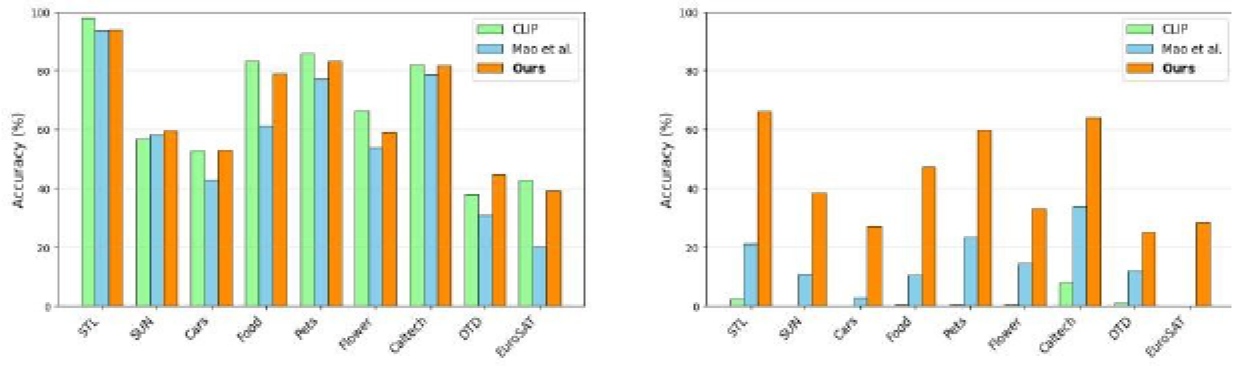
(a) Denoised smoothing from text-to-image models (b) Self-adaptation schemes

- Unsupervised calibration: entropy minimization
  - Feature-stat EMA for test-time stabilization
  - Prompt and augmentation ensembling boosts robustness
- Requires no labeled data; fully test-time adaptive
  - Classifier-guided regularization aligns denoised outputs
  - Complements denoised smoothing for stronger robustness

Sec. 3

Experiments: CLIP & Classifiers

↑ Evaluation of CLIP



(a) Clean accuracy (b) Robust accuracy

- Surpasses randomized smoothing and adv. fine-tuning
  - Large robust gains with minimal clean accuracy drop
  - Effective across zero-shot and domain-specific datasets
- Consistent robustness gains without external data

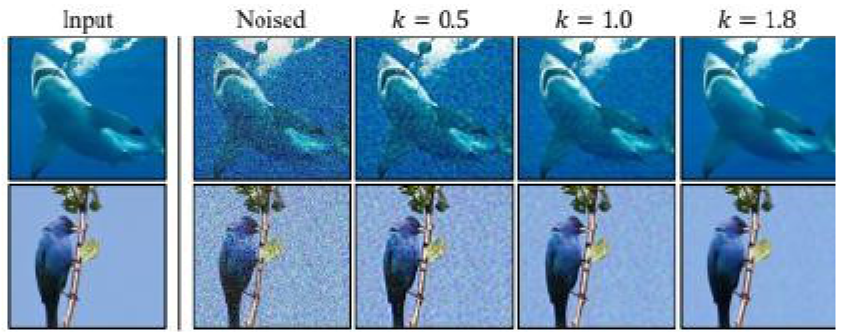
9 Generic Vision Classifiers

- Evaluated on ImageNet with ResNet-50 and ViT models.
- Denoised smoothing plus self-adaptation boosts robustness.
- Consistent gains achieved without any model retraining.
  - Outperforms adversarial training and data-dependent baselines.
  - Scalable and effective across diverse application scenarios.
- Robust benefits transfer across standard classifier families.

Sec 10

Ablation Study

11 Impact of Design Choices



- Moderate noise k best balances purification and detail
  - More samples stronger denoising boost robustness at cost
  - Adaptation steps help early, with diminishing returns
- Timestep correction factor is key to denoising quality
  - Visual series shows purification-detail trade-off
  - Adaptation schemes contribute to robustness, details vary

12 Conclusion and Implications

- Training-data-free framework with provable robustness.
- Leverages text-to-image diffusion, denoised smoothing, self-adapt.
- Delivers strong robustness while preserving clean accuracy.
  - Off-the-shelf classifiers gain provable robustness; minimal data.
  - Enables deployment on private or unavailable datasets.
- Future: certified analysis, faster diffusion, broader modalities.